

Amrita Vishwa Vidyapeetham - Amrita School of Engineering- Coimbatore
Department of Electronics and Communication Engineering

Acad. Year	Semester	Duration	Class	Batch
2026-27	Odd	June 2026 – October 2026	5 Sem ECE	2024-2028
23ECE381 Open Laboratory I - Review ZERO: Proposal Submission*				

Batch Number:	1
Project Title:	Acoustic phased-array system
Review Date:	

Members:		
S.No	Name	Roll Number
1.	Beesetty Rohith	CB.EN.U4ECE24106
2	Indraganti Sri Sarath	CB.EN.U4ECE24122
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Description of Work
The idea behind a phased-array is that by changing how the speakers are driven the angle of the maximum intensity of the output wave can be shifted. This type of array was built to be able to support various other more advanced design challenges, including longer-range acoustic modem transmission and sonar imaging. Building a wide-band phased array required true time delay between the elements, a task that a microcontroller is particularly well-suited for. The physics of a phased-array of transmitters can be derived from the math for a diffraction grating; In the far-field, the wave function of such a system is; $\Psi = \Psi_0 \frac{\sin(\frac{\pi a}{\lambda} \sin \theta)}{\frac{\pi a}{\lambda} \sin \theta} \frac{\sin(\frac{N}{2} \frac{2\pi d}{\lambda} \sin \theta)}{\sin(\frac{\pi d}{\lambda} \sin \theta)}$ The first fraction is just the diffraction pattern of one speaker of width a (same shape as single-slit diffraction, a broad central bump). The second fraction is the interference pattern you get from combining N identical speakers spaced d apart (same shape as a diffraction grating – narrow sharp peaks), this is the part that actually matters for beam-steering. When we add the phase term, Φ, to the second part of the equation, it becomes; $\Psi = \Psi_0 \frac{\sin(\frac{\pi a}{\lambda} \sin \theta)}{\frac{\pi a}{\lambda} \sin \theta} \frac{\sin(\frac{N}{2} (\frac{2\pi d}{\lambda} \sin \theta + \Phi))}{\sin(\frac{\pi d}{\lambda} \sin \theta + \frac{\Phi}{2})}$ Next, we square the wave function to get the intensity of the wave;

$$I = I_0 \left(\frac{\sin\left(\frac{\pi a}{\lambda} \sin \theta\right)}{\frac{\pi a}{\lambda} \sin \theta} \right)^2 \left(\frac{\sin\left(\frac{N}{2} \left(\frac{2\pi d}{\lambda} \sin \theta + \Phi\right)\right)}{\sin\left(\frac{\pi d}{\lambda} \sin \theta + \frac{\Phi}{2}\right)} \right)^2$$

To find the steering angle, or to find the main beam direction the second fraction shaped like $\sin(Nx)/\sin(x)$ peaks when the denominators argument goes to zero, then solving for θ in terms of Φ , gives;

$$\begin{aligned} \sin\left(\frac{\pi d}{\lambda} \sin \theta + \frac{\Phi}{2}\right) &= 0 \\ \frac{\pi d}{\lambda} \sin \theta + \frac{\Phi}{2} &= 0 \\ \theta &= \sin^{-1}\left(\frac{-\lambda}{2\pi d} \Phi\right) \end{aligned}$$

Thus, we can see that we should expect the angle of maximum amplitude to be a function of both the phase shift of the wave, and its wavelength (assuming the distance between the sources is fixed). For microcontroller coding, we would prefer to not require different calculations for each frequency. Fortunately, some mathematical manipulation shows us an easier method of adjusting the angle for all frequencies at once.

We Know That, $\lambda f = v_s$; then,

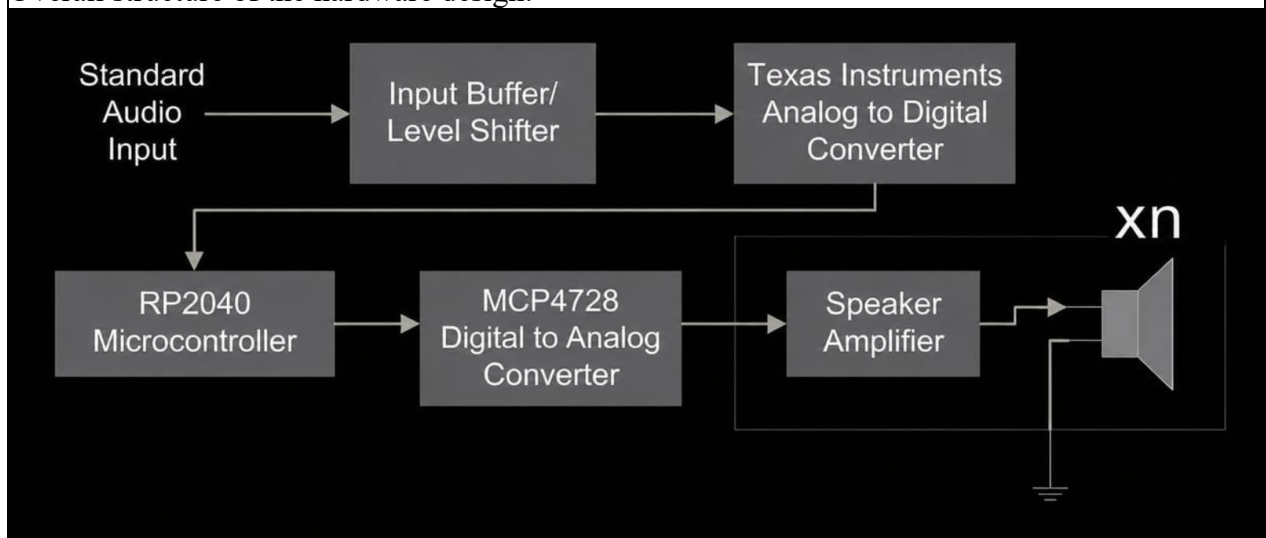
$$\theta = \sin^{-1}\left(\frac{-v_s}{2\pi d} \frac{\Phi}{f}\right)$$

We recognize that for any frequency, $\frac{\Phi}{f}$ is equivalent to a time delay. Thus we get

$$\theta = \sin^{-1}\left(\frac{-v_s}{2\pi d} t_d\right)$$

Now we have an equation that relates the maximum power angle to delay time between the elements. This equation has no dependence on wavelength, indicating it should hold true for a large band of frequencies. (This part will be verified using MATLAB simulation.)

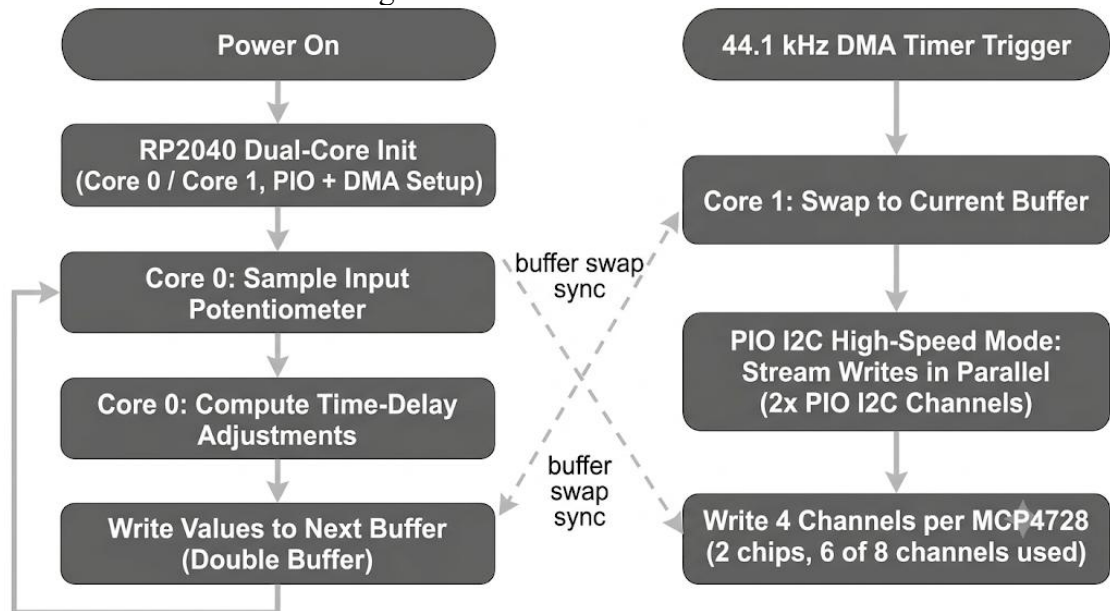
Overall structure of the hardware design:



Our input is taken from any standard audio source. A general inspection of the signal is done to check if it is varying between plus and minus 1V. To get the full use of our Analog to Digital converter (ADC), we needed a signal that varies between 0 and 5V. Therefore, we use an input

amplifier to re-bias and amplify the signal before going into the ADC. The ADC then communicates the digital value of this input signal to the RP2040 microcontroller. The microcontroller performs the necessary mathematics for the desired phase shift between the speakers, and then outputs each channel to a Digital to Analog converter (DAC). This output is then passed to a speaker amplifier, which both applies a low-pass filter to the signal and buffers it before outputting it to the speaker. The low-pass filter is necessary because the use of DACs introduces quantization noise, and the buffer is necessary because the impedance of the speaker is only 8 ohms, much lower than the output impedance of the DAC.

Overall structure of the software design:



Upon power-up the microcontroller initializes its hardware. This includes setting up SPI communication with the ADC, setting up the internal ADC, and setting up the timer interrupts to ensure we get sound capture and play-back at a well-defined resolution. Once this hardware is initialized, the main loop of the program runs. This is fairly simple, it reads an input channel connected to a potentiometer which acts as the user interface, and then sets the delay for the desired angle accordingly. This main loop is interrupted at a frequency of 44.1 kHz to perform audio capture and playback. In the interrupt, the time-delayed outputs are sent to each channel of the DACs, and the new input data is sampled from the ADC.

Hardware Justifications:

1. Microcontroller:

The one function a microcontroller and DAC is doing here is converting a live, continuously-adjustable steering angle per-channel time delays applied to a live sampled waveform is inherently an arithmetic/sequencing task, not a fixed signal conditioning task.

The discrete-hardware alternative would be something like a bank of analog bucket-brigade delay-line ICs (e.g. BBD chips) with separate clocking per tap. This too doesn't actually increase the hardware design content, but just swapping one IC based subsystem for another, while losing quality (BBDs add their own companding noise and clock-feedthrough distortion, working against the speaker quality goal.)

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2. Speaker:				
Parameter	Dayton Audio CE36-8	PUI Audio AS03208MS-3-R	Challenge Electronics CS32-03W23-16-1X	PUI Audio AS07108PO-3-R
Impedance	8Ω	8Ω	8Ω	8Ω
Power	3W / 6W	3W / 4W	3W / 4W	3W / 5W
Sensitivity	81dB @ 2.83V/1m	85dB	83dB	86dB
Frequency Range	200–19,000 Hz	200 Hz–20 kHz	184 Hz–20 kHz	100 Hz – 20 kHz
Resonant Frequency	200 Hz	-	230 Hz	250 Hz
Distortion Rating	-	-	10% max @ 1 kHz, 1W	-
Size	1" round	1.25" round	1.29" round	71x41mm oval
Documentation	Full T-S parameter set	Full datasheet	Full datasheet	Full datasheet
Price	-	700 – 900/-	550 – 750/-	400 – 500/-
The PUI Audio AS07108PO-3-R was selected as it is the only candidate that meets the 86dB sensitivity target exactly. It is manufactured by PUI Audio, a reputable audio component supplier with full datasheet documentation. Its extended low-frequency response also gives the array better full-range reproduction.				
References: (Preferably in IEEE Format)				
Y. Luo, "Constant Directivity Loudspeaker Beamforming," 2024. arXiv:2407.01860v3. C. Pope, H. Tang, B. Zheng and H. Zhang, "Phased Array Systems – Design Considerations & System Demonstration," 2024 IEEE International Symposium on Phased Array Systems and Technology (ARRAY), Boston, MA, USA, 2024, pp. 1-8, doi: 10.1109/ARRAY58370.2024.10880346. B. Rafaely and D. Khaykin, "Optimal model-based beamforming and independent steering for spherical loudspeaker arrays," <i>IEEE Transactions on Audio, Speech, and Language Processing</i>, 2011. arXiv:2310.04202. Zhuang, T., Zhong, J.-X., & Lu, J. (2024). The feasibility of sound zone control using an array of parametric array loudspeakers. <i>arXiv preprint arXiv:2407.10054</i>. D. de Groot, B. Karslioglu, O. Scharenborg, and J. Martinez, "Loudspeaker Beamforming to Enhance Speech Recognition Performance of Voice Driven Applications," arXiv:2501.08104 [eess.AS], Jan. 2025. Hardware reference: A. Da Silva Gaviola, M. Rivai and H. Kusuma, "Audio beam steering with phased array				

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method using Arduino Due Microcontroller," *2018 International Conference on Information and Communications Technology (ICOIACT)*, Yogyakarta, Indonesia, 2018, pp. 597-600, doi: 10.1109/ICOIACT.2018.8350683.

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Components Required: (Hardware and Software (EDA))					
S.No	Product Name	Specification	Quantity	Unit Price (Rs.)	Available in Lab/ Campus / Put tick if available
1	Bread Board	-	1/2	-	✓
2	Microcontroller	RP2040	1	80 – 100/-	
3	Speaker	8Ω 3W 86dB	6	100 – 500/-	
4	Transistor	NPN	12	-	✓
5	Mosfet	NMOS	12	-	✓
6	Opamps	Qaud Op-amp	3	-	✓
7	DAC	MCP4728	2	180 – 250/-	
8	Potentiometer	10k	1	-	✓
9	Resistors	As required	As required	-	✓
10	Connecting Wires	-	As required	-	✓
11	Capacitors	As required	As required	-	✓
12	Audio Jack	3.5mm	1	150/-	
13	Function Generator	3 MHz	1	-	✓
14	Power Supply	0 – 30V	1	-	✓
15	Probes	-	As required	-	✓

Reviewer Comments:

Status of Proposal: Put tick in relevant column

ACCEPT:	REJECT:

* The same proposal to be converted as a product in Open Laboratory II
 #Approval from faculty required

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Reviewer Signatures (with Date)	
Name:	Name: